

S5 - Glossary

Shared terminology across articles

Short title: Glossary - NeuralManifoldDynamics

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Abstract

This document defines the core terminology used across articles in this repository. Entries cover the theoretical framework, the computational pipeline, embodied anchor metrics, the family of primary metric scalar summaries, trajectory geometry, robustness-audit methods, and statistical methods. Definitions reflect the current implementation as of the latest pipeline version; version-specific changes to metric computation are noted in the relevant entry.

This is a living glossary. “Implemented” describes the computation or export surface; “interpretation” describes the narrowest defensible reading; and “not established” identifies a theoretical or clinical claim that the current repository does not prove.

A - D

Attractor - Not established in this ingest contract. In dynamical-systems language, a stable region to which trajectories return after perturbation. NeuralManifoldDynamics exports trajectories and local summaries; it does not identify attractors.

Anchor state - Implemented. A time-aligned physiological feature surface, including HRV-, PPG-, or pupil-derived indices. It is additive and does not replace `mnp3d`.

A_{BD} - Operational alignment diagnostic. The scalar $b^T \hat{a} / (b^T b + \epsilon)$ used when an independently qualified chart velocity b is available. It is not testable without that velocity and is not an Itô drift-to-diffusion identification.

Anisotropy - Implemented. A summary of directional imbalance in a Jacobian or block-Jacobian field. Its reliability depends on local support and conditioning.

CHART - Bounded coordinate audit. A synthetic audit of coordinate and projection behavior. Orthogonal and isotropic cases pass the audited identity; anisotropic cases are expected to change local estimates, projection loses hidden-coordinate information, and the scalar unregularized diagnostic is not a production C1 gate.

Claim status - Implemented provenance field. A machine-readable ceiling on interpretation, including `no_biological_claim` for computed dynamical-family objects.

Coordinate - Implemented. A release-fixed numerical axis or subcoordinate. The labels `m`, `d`, and `e` are operational proxy families, not uniquely identified biological quantities.

Coverage - Implemented. The fraction of configured feature or axis support available after masking and quality control. Minimum coverage is part of geometry validity.

Derived layer - Implemented. An event-locked or block-native sidecar that joins to canonical windows through provenance but does not redefine the HDF5 measurement contract.

E - L**Embodied Anchoring Principle (EAP) - Implemented as an additive design principle.**

Body-state measurements share the MNPS time grid while remaining separate from the canonical neural chart.

Event-locked - Implemented derived layer. A window table aligned to an event onset or peak, with detector and control-matching provenance.

Feature anchor - Implemented. Frozen per-feature center/scale metadata used to construct cohort-anchored coordinates. It is distinct from `anchor_state`.

Feature standardization - Implemented. The configured transform applied before projection, commonly robust-z plus clipping; selected power features may use log10 first. The transform is recorded for audit.

Frobenius norm - Implemented. The square root of the summed squared Jacobian entries; a magnitude summary, not a causal strength.

Geometry contract - Implemented. The set of support, finite-value, conditioning, duplicate-coordinate, and time-grid rules used to decide whether a trajectory or Jacobian is valid.

Geometry policy standard_invalid_v1 - Implemented in v2.4.0. The versioned policy that records canonical invalidity and `coords_9d` duplicate tolerance diagnostics.

HRV v0.1 - Implemented in v2.4.0. A centered ECG superwindow surface with HR mean, IBI mean, SDNN, RMSSD, pNN50, NN count, artifact fraction, coverage fraction, and quality score. HF and LF/HF are not included.

Jacobian / MNJ - Implemented optional summary. A local estimate of $\partial f / \partial x$ in chart coordinates. It is not a direct estimate of a unique biophysical state equation.

Analytic conditional-lag witness - Model-derived diagnostic. For a linear Gaussian system, the Frobenius norm of $C_2 - C V^{-1} C$ measures one-step lag memory after conditioning

on the current observation, where $C_2 = H A^2 P$

H^T . The conditional next-state covariance is a separate quantity, $V = C V^{-1} C^T$. Neither is an estimator variance, Markov-sufficiency gate, or biological qualification.

Cross-segment order corruption - Synthetic negative control. A fixed roll of held-out predictions changes the source-to-truth mapping while preserving the truth array. It is a mapping control, not a temporal-shift estimator or a biological null.

Discrete conditional velocity - Analysis-owned target. A one-step conditional expectation divided by the declared sampling interval. It is a discrete chart target and does not assert an Itô drift.

Dynamical family - Implemented gated surface. A versioned measurement family beside the canonical chart, such as diffusion, destination, or resilience. It carries support, validity, grain, provenance, and claim metadata and does not add canonical MNPS axes.

M - R

MEG shadow mapping - Implemented exploratory path. `meg_*` features route through the existing 9D projection slots used by the EEG-family contract. Cross-modal biological equivalence is not established.

Neuropixels Units-to-rate - Implemented in v2.5.0, transport/descriptive status. NWB spike times are histogram-binned (default 0.05 s bins, optional 0.05 s Gaussian smoothing) into a population firing-rate matrix, then windowed into `ephys_*` features that fill `coords_9d` through a dedicated ephys-specific subcoordinate map. Validated as a one-session smoke run on a rodent dandiset; not a validated manifold claim.

Neuropixels LFP-as-EEG routing - Implemented in v2.5.0, transport/descriptive status. DANDI ElectricalSeries LFP channels are routed through the existing EEG feature and 9D

contract with rodent-adapted band edges and optional depth-balanced contact selection. Validated through smoke and full-session QC on a rodent dandiset; no readiness score is computed.

Measurement contract - Implemented. The versioned definition of what is computed, named, quality-controlled, and serialized for downstream reuse.

MNPS / mnps_3d - Implemented operational chart. The canonical three-axis trajectory $[m, d, e]$ exported by the current contract.

Regional MNPS - Implemented optional output. Separate trajectories and Jacobians for fMRI networks or EEG channel groups, subject to modality-specific support limits.

Measurement validity - Implemented certificate field. A declared validity state for a computed object, such as `not_assessed`, `insufficient_support`, or `translation_qualified` where the named protocol permits it. It is distinct from `validation_level` and from `claim_status`.

Grain - Implemented schema metadata. The native row and biological unit of an output, including recording, subject, and window context. Windows are not biological replicates and direct between-subject inference is forbidden on these family exports.

Support signature - Implemented capability metadata. A machine-readable summary of coordinate support, modality capability, and fallback provenance; it does not infer validity or scientific licensing from the presence of a dataset.

W_Q - Implemented opt-in discrete reachability object. A chart-space one-step predictive-spread covariance under the Gate E recording-level transition-residual contract, written under `/stochastic_reachability/v1`. It is not a $\hat{a}$, Itô process noise, or a default dynamical-family output.

Readiness score - Implemented pilot gate. A weighted engineering/validation summary, not a clinical or biological probability. The ds003645 pilot score was 0.7879.

Row provenance (row_source) - Implemented in v2.4.0. Per-window source metadata for simultaneous MEEG rows, avoiding an implicit positional split.

Robust-z guard - Post-v2.4 working-tree safeguard. A policy for degenerate feature scales that can emit NaN rather than misleading extreme values. It is not part of the frozen v2.4.0 contract until versioned.

S - Z

Subject-anchored coordinates - Implemented. Coordinates preserving within-subject or within-session geometry.

Cohort-anchored coordinates - Implemented. Coordinates based on a frozen external or fitted feature anchor for between-subject comparison.

features_projection_z - Implemented in v2.4.0. A transform-aware feature surface used when physical-scale features, especially MEG power, require log10 before robust-z and clipping.

Block-native - Implemented derived layer. Sustained-block windows with block labels, relative position, and source_window_index; appropriate for slower variables such as HRV.

BH-FDR - Implemented where declared by a validation package. Benjamini–Hochberg correction controls a named multiple-comparison family. The current methods article does not define one global coordinate FDR family.

LOOSO - Implemented in selected analyses. Leave-one-subject-out recalculation used for cohort-specific robustness, not a universal property of all outputs.

Bootstrap - Implemented in selected follow-ups. Resampling used to characterize uncertainty in a named validation package.

Permutation / label-shuffle null - Implemented in selected analyses. Labels or pairings are shuffled to form a stated null; the null design and unit of exchangeability must be reported with the result.

Phase anchor - Implemented optional v2.5.0 surface. An optional per-epoch extractor for cardiac phase, respiratory phase, RR intervals, HR, respiration, and HEP-related features. It was absent from the frozen v2.4.0 contract and remains dataset-dependent.

Proxy - Interpretation boundary. A reproducible computational correlate used under a contract; it is not automatically a latent state, biomarker, or clinical endpoint.

Observed conditional covariance - Implemented analysis quantity. The analytic conditional covariance for an observed one-step target, retained separately from the raw local increment covariance. It is not a substitute for the production diffusion object.

Pointwise MCSE - Descriptive uncertainty summary. The standard error of a prediction or alignment mean across the twelve independent training fits at the same held-out query, computed as sample standard deviation divided by $\sqrt{12}$. No standard error of the variance is reported. It is not a confidence interval for a biological effect.

Raw increment covariance - Implemented diffusion quantity. The local covariance of raw increments divided by the declared sampling interval. A fixed evaluation-neighbour set is a property of the bounded M3 benchmark, not a universal production rule. This quantity remains distinct from an analytic conditional covariance and from an unobserved latent drift.

Bibliography

Competing Interests

The authors have declared that no competing interests exist.